SYNOPSIS

1. INTRODUCTION:

Indoor air quality (IAQ) is an important environmental factor that directly influences human health, comfort, and productivity. While much attention is often given to outdoor pollution, indoor environments, such as classrooms, hostels, laboratories, and offices, can also experience significant deterioration in air quality. These spaces are typically enclosed, frequently occupied, and depend heavily on natural or mechanical ventilation. As a result, pollutants such as carbon dioxide (CO_2), particulate matter (PM2.5), volatile organic compounds (VOCs), and humidity fluctuations can accumulate over time without being noticed. Continuous exposure to such conditions can lead to fatigue, headaches, irritation, loss of concentration, and long-term respiratory issues. This creates a need for systematic monitoring of indoor air, especially in educational institutions where students spend long hours in confined spaces.

In recent years, the increasing use of electronic devices, cleaning chemicals, and synthetic materials has further contributed to indoor air contamination. Traditional buildings are often designed without adequate ventilation provisions, and modern energy-efficient buildings tend to restrict natural airflow. This lack of ventilation allows pollutants to build up gradually, making indoor air potentially more harmful than outdoor air. Research studies have shown that high CO_2 levels in crowded classrooms can negatively impact attention span and learning efficiency. Similarly, elevated levels of particulate matter and VOCs have been linked to allergies, breathing difficulties, and general discomfort among occupants. A simple visual inspection cannot detect these pollutants, making real-time monitoring essential for maintaining a healthy indoor environment.

With the advancement of sensor technology and Internet of Things (IoT) systems, it is now possible to develop low-cost, reliable, and user-friendly platforms for monitoring air quality. These systems can continuously measure pollutant levels, visualize data on dashboards, and generate alerts when values exceed safe limits. For educational institutions and hostels, such systems offer an effective way to ensure that students and staff are protected from poor indoor conditions. By understanding air quality patterns, management can take timely actions such as increasing ventilation, reducing occupancy, or adjusting schedules to improve comfort and well-being.

The present project, **AirShield**, aims to design and implement an IoT-based indoor air quality monitoring and alert system that measures key environmental parameters in real time. The system provides a practical and affordable solution that supports better decision-making,

encourages cleaner indoor environments, and promotes overall health and productivity. This technology-driven approach aligns with current needs in environmental engineering and offers a sustainable method for maintaining safe indoor spaces in academic and residential settings.

1.1 Importance of Indoor Air Quality:

Indoor air quality plays an essential role in ensuring safe and healthy living and working conditions. Poor IAQ has been linked to Sick Building Syndrome (SBS), reduced cognitive performance, increased absenteeism, and long-term respiratory complications. In educational institutions, where students spend long hours in classrooms and hostels, maintaining optimal air quality becomes crucial for better learning outcomes. Elevated CO_2 levels can slow response times, reduce concentration, and impair decision-making abilities.

Additionally, particulate matter and VOC exposure can aggravate asthma, allergies, and other respiratory conditions, particularly affecting children and young adults. Monitoring IAQ allows for early identification of pollution sources, promotes better ventilation practices, and ensures compliance with recommended environmental health standards. A real-time monitoring system empowers users to make informed decisions, take corrective measures, and maintain healthier indoor environments.

1.2 Possible Ways to improve Indoor Air Quality

i. Improving natural ventilation inside the room by opening windows and allowing fresh air circulation.
ii. Minimizing indoor pollution sources such as dust, chemicals, and improper waste handling.
iii. Maintaining cleanliness and regular housekeeping to reduce particulate matter.
iv. Using basic ventilation aids like exhaust fans or simple air-moving devices where required.
v. Monitoring indoor conditions regularly to identify deterioration in air quality and take timely corrective actions.

2. REVIEW OF LITERATURE:

Research on indoor air quality monitoring and assessment has been carried out using various analytical and sensor-based approaches. Some relevant studies include:

Peavy, Rowe and Tchobanoglous (1985) explain the fundamental principles of environmental pollution and its impact on human health. Their discussion on indoor air contaminants, pollutant dispersion, and ventilation requirements provides a scientific basis for selecting IAQ parameters. The book, *Environmental Engineering* (McGraw-Hill), emphasizes the need for continuous monitoring in enclosed spaces where pollutant accumulation can occur rapidly during regular occupancy.

Masters and Ela (2008) describe the chemistry and behavior of air pollutants, including particulate matter, CO_2, and VOCs, and their effects on human comfort and cognitive performance. They highlight the importance of assessing indoor environments where extended human presence can lead to pollutant buildup in their book *Introduction to Environmental Engineering and Science* (Pearson). Their work supports the requirement for low-cost, practical monitoring systems in educational institutions.

Rao M. N and Rao H. V. N (2001) present detailed insights into the sources, characteristics, and health impacts of air pollutants. They highlight that indoor concentrations of particulate matter and gases may exceed outdoor levels due to inadequate ventilation. Their literature, *Air Pollution* (Tata McGraw-Hill), stresses that regular monitoring is necessary to identify unsafe indoor conditions, especially where occupancy is high.

Wark, Warner and Davis (1997) discuss air pollution origins, threshold limits, and measurement instrumentation used for pollutant detection in the book *Air Pollution: Its Origin and Control* (Harper & Row). They emphasize that accurate measurement is essential before any control measures can be implemented. This aligns with the objective of developing a monitoring-first system like AirShield.

Stern A. C. (1977), in *Air Pollution* (Academic Press), focuses on air quality measurement techniques and the challenges associated with detecting low-level pollutants indoors. He explains the limitations of manual sampling and advocates for automated, continuous

monitoring systems. This reinforces the need for real-time sensing in small indoor environments such as classrooms and hostels.

Wei et al. (2021) review machine learning and statistical models for predicting indoor air quality using sensor data. Their study highlights rapid pollutant fluctuations in occupied rooms and the importance of continuous monitoring to capture such variations. This literature supports incorporating real-time IAQ tracking in the proposed system.

Sharma and Kumar (2020) analyze various low-cost environmental sensors and their applicability to indoor air monitoring. They conclude that these sensors can produce reliable data when properly calibrated. Their work validates the feasibility of using affordable sensor-based systems for indoor environments.

IndoAirSense Team (2021) proposes a structured IAQ forecasting framework for classrooms and offices. Their findings show strong correlations between occupancy patterns and pollutant levels, indicating the need for uninterrupted monitoring. This supports the deployment of sensor-based IAQ systems in educational settings.

World Health Organization (2021) provides guideline values for major pollutants such as PM2.5, CO_2, and VOCs, defining acceptable and hazardous exposure levels. These guidelines serve as reference thresholds for alert generation in IAQ monitoring systems and are used to establish safety indicators in this project.

Neptronic (2022) provided technical documentation related to commercial IAQ solutions used in large buildings. Although not academic research, it presents practical industry standards for pollutant thresholds and the functionalities expected in modern air monitoring systems.

These studies collectively highlight the necessity for a low-cost, real-time, IoT-based indoor air quality monitoring system, validating the relevance of the proposed **AirShield** project.

3. PROBLEM STATEMENT:

Indoor air quality in enclosed spaces such as classrooms and hostels often deteriorates due to inadequate ventilation and continuous occupancy. Pollutants such as CO_2, particulate matter, VOCs, and humidity variations accumulate over time without being noticed, leading to discomfort, reduced concentration, and potential health issues. Most educational buildings do not employ continuous monitoring systems, resulting in delayed identification of poor indoor conditions. Existing commercial solutions are expensive and not suitable for small institutional settings. Hence, there is a need for a simple, low-cost system capable of monitoring essential indoor air quality parameters and indicating unsafe conditions in real time.

4. SCOPE OF THE PROJECT:

The present study focuses on the design and development of an IoT-based indoor air quality monitoring system suitable for small, enclosed environments such as classrooms, hostels, and laboratories. The system will measure selected parameters including CO_2, PM2.5, VOCs, temperature, and humidity, and present the collected data through a real-time dashboard along with basic alert indications. The scope is limited to monitoring and interpretation of air quality and does not include the design or control of ventilation equipment, air purifiers, or building management systems. The study covers sensor integration, data transmission, cloud storage, and dashboard visualization within the defined indoor settings.

5. OBJECTIVES:

Following will be the objectives of proposed project:

i. To monitor selected indoor air quality parameters such as CO_2, PM2.5, VOCs, temperature, and humidity.

ii. To develop a simple IoT-based system for recording and displaying real-time indoor air quality data.

iii. To provide basic alert indications when pollutant levels exceed recommended limits.

iv. To create a user-friendly dashboard for interpreting variations in indoor air quality.

v. To promote awareness regarding the need for maintaining healthy indoor environments.

6. METHODOLOGY

i. **Parameter Selection:** Finalized IAQ parameters (CO_2, PM2.5, VOCs, temperature, humidity) using literature and WHO guidelines.

ii. **Hardware Selection:** Selected suitable sensors and a Wi-Fi microcontroller (ESP8266/ESP32).

iii. **Sensor Integration:** Connected and mounted sensors onto a prototype board; tested basic functionality.

iv. **Firmware Development:** Programmed the system to read, process, and transmit sensor data.

v. **Cloud Setup:** Configured IoT cloud channels (ThingSpeak/Firebase) for data storage and visualization.

vi. **Dashboard Design:** Created a real-time dashboard with graphs and color-coded indicators.

vii. **Alert System:** Set threshold values and implemented alerts via indicators/buzzer/ notifications.

viii. **Field Testing:** Deployed the system indoors and compared results with standard limits.

7. FACILITIES AVAILABLE:

Following are the facilities available to complete the project:

i. Departmental and central library resources for literature and reference material.

ii. Environmental Engineering and IoT laboratory facilities for testing and setup.

iii. Computer systems with internet access for programming, cloud integration, and dashboard development.

8. TENTATIVE EXPENDITURE EXPECTED:

The expected expenditure for the project is **Rs. 5,000 to Rs. 8,000**, depending on sensor selection and hardware requirements.

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